

Experiment 5 : Comparison of Naïve and Basic Prompt Patterns

Aim

To evaluate and compare how different prompt pattern styles influence AI responses by testing unstructured (naïve) prompts versus well-defined (basic) prompts across multiple use cases. The comparison focuses on response quality, correctness, clarity, and level of detail.

AI Tool Required

ChatGPT

Explanation

Naïve prompts are loosely framed prompts with minimal context and vague instructions. Basic prompts are structured and detailed prompts that provide context, constraints, and expected output format. Multiple scenarios are tested using both prompt types to compare their effectiveness.

OUTPUT

Scenario 1: Creative Content Generation

Naïve Prompt: Write something about artificial intelligence.

Output: Artificial intelligence is widely used today. It helps people in many areas such as education, business, and healthcare.

Basic Prompt: Write a short creative story about an AI assistant in a hospital that helps doctors during emergencies.

Output: In a busy hospital, an AI assistant named MediCore monitored patient data in real time and alerted doctors during emergencies, helping save lives.

Scenario 2: Factual Explanation

Naïve Prompt: What is cybersecurity?

Output: Cybersecurity is about protecting computers and data from attacks.

Basic Prompt: Define cybersecurity and explain its importance with one real-world example.

Output: Cybersecurity protects systems and data from digital threats. For example, antivirus software prevents malware attacks.

Scenario 3: Concept Summarization

Naïve Prompt: Explain electric vehicles.

Output: Electric vehicles run on electricity and are eco-friendly.

Basic Prompt: Summarize electric vehicles in 5–6 lines covering benefits and impact.

Output: Electric vehicles use batteries instead of fuel, reduce pollution, lower operating costs, and support sustainable transportation.

Scenario 4: Advice and Recommendations

Naïve Prompt: Give health tips.

Output: Health can be improved by eating well and exercising.

Basic Prompt: As a health advisor, suggest three daily habits to maintain good health.

Output: Balanced diet, regular exercise, and proper sleep help maintain good health.

Observation Summary

Naïve prompts produced average-quality responses with low depth, while basic prompts generated accurate, well-structured, and detailed responses.

Result

The experiment confirms that well-structured basic prompts significantly improve the quality and effectiveness of AI-generated responses compared to naïve prompts.